# Evolutionary Antecedents of of Motivation

A University Lecture Series

Department of Behavioral Sciences

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# The Enduring Question of Human Nature

The concept of a shared 'human nature' has been a subject of contemplation for centuries, posing fundamental questions about the core of our being.

> *"Human nature is the same all over the world, but its operations are so varied by education and habit that one must see it in all its dresses in order to be entirely acquainted with it."*
>
> — Lord Chesterfield, 1747

> *"Human action can be modified to some extent, but human nature cannot be changed."*
>
> — Abraham Lincoln, c. 1860

These observations force us to ask: Does human nature refer to motivational characteristics that all people have in common, despite vast cultural differences? Is there a universal core of motives and goals that defines our species?

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# Defining Human Nature from an Evolutionary Standpoint

From an evolutionary perspective, human nature is not a vague philosophical idea but a concrete scientific concept.

It refers to the collection of behavioral, motivational, and emotional similarities among all people that have resulted from our common evolutionary history. These are not rigid commands but rather dispositions to behave in particular ways depending on the situation.

Human nature is most striking when we observe these profound similarities occurring in spite of vastly different environmental and cultural contexts. Its key characteristics are:

- It is shaped by natural and sexual selection.
- It is genetically transmitted across generations.
- It is universal.
- Its behavioral expressions tend to be innate, meaning they are influenced little by experience.

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# Evolutionary Perspective

## The Ultimate Goals: Survival and Gene Transmission

From an evolutionary standpoint, the fundamental purpose of motivation is elegantly simple: it pushes and pulls individuals toward actions that facilitate the survival and transmission of their genes to the next generation.

This overarching goal breaks down into a series of crucial life tasks that have been repeated for millennia (Bernard et al., 2005):

- Surviving to reproductive age.
- Finding and keeping a suitable mate.
- Producing viable offspring.
- Raising those offspring successfully so they can repeat the process.

This relentless cycle, played out over deep time, has sculpted the tfe set of common behaviors we identify as human nature.

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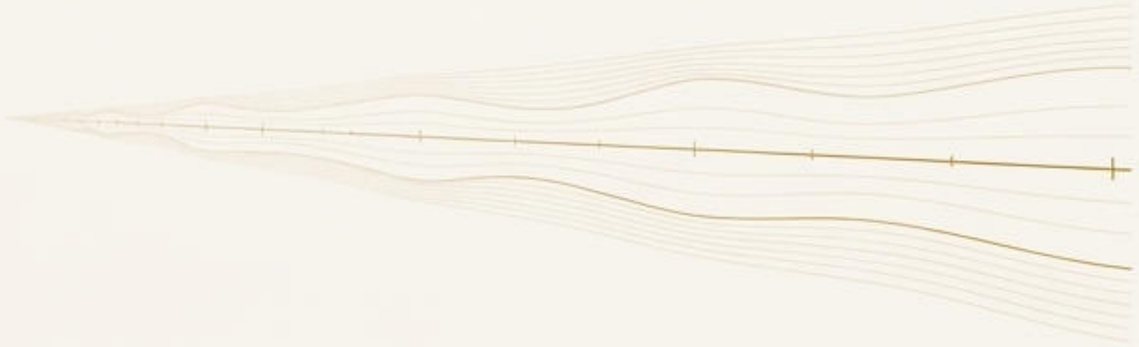
# A Tale of Two Histories

To understand the motivation for any current behavior, we must consider its origins. Where in an individual's past do we locate the cause? This creates a fundamental distinction between two timelines:

## 1. Evolutionary History

The remote past of our species, encompassing the long-term selective pressures that shaped our genetic inheritance. This is the ultimate, or "first," cause.



## 2. Personal History

The recent past of an individual, comprising their unique environmental, cultural, and learning experiences. This is the proximate, or "immediate," cause.

The central question becomes: How do these two histories—our deep evolutionary past and our immediate personal experiences—combine to determine what is motivating for us in the present moment?

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# The Domino Metaphor: A Causal Chain

Imagine a line of dominoes. The fall of the final domino—representing our current motivated behavior—is the end result of a long chain reaction. Is the cause the fall of the immediately preceding domino (personal history)? Or is the ultimate cause the toppling of the very first domino (evolutionary history)?

The answer is both. Evolutionary history is the initial event that topples the first dominoes. This sets in motion a sequence where later dominoes, representing personal history, are toppled. The fall of the later dominoes is entirely dependent on the fall of the earlier ones. Our current motivation is the final domino, a product of this entire, interconnected sequence.

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# Nature & Nurture Interaction

## Beyond Dichotomy to Interactionism

The relationship between evolutionary history (nature) and personal history (nurture) is one of interaction, not opposition. This idea replaces the outdated ‘nature versus nurture’ debate.

Sir Francis Galton recognized this complexity over a century ago, stating in 1883 that it was difficult to disentangle the parts of human character resulting from “education and circumstances” from that which results from our innate “constitution.”

Contemporary psychology confirms this view: it is an established principle that heredity and environment dynamically interact to determine behavior (Plomin et al., 2013). They are not separate forces but interwoven contributors.

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# The Rectangle Metaphor: A Model of Interaction

We can visualize the interaction between heredity and environment using the analogy of a rectangle's area (Plomin et al., 2013). The area of a rectangle is calculated as:

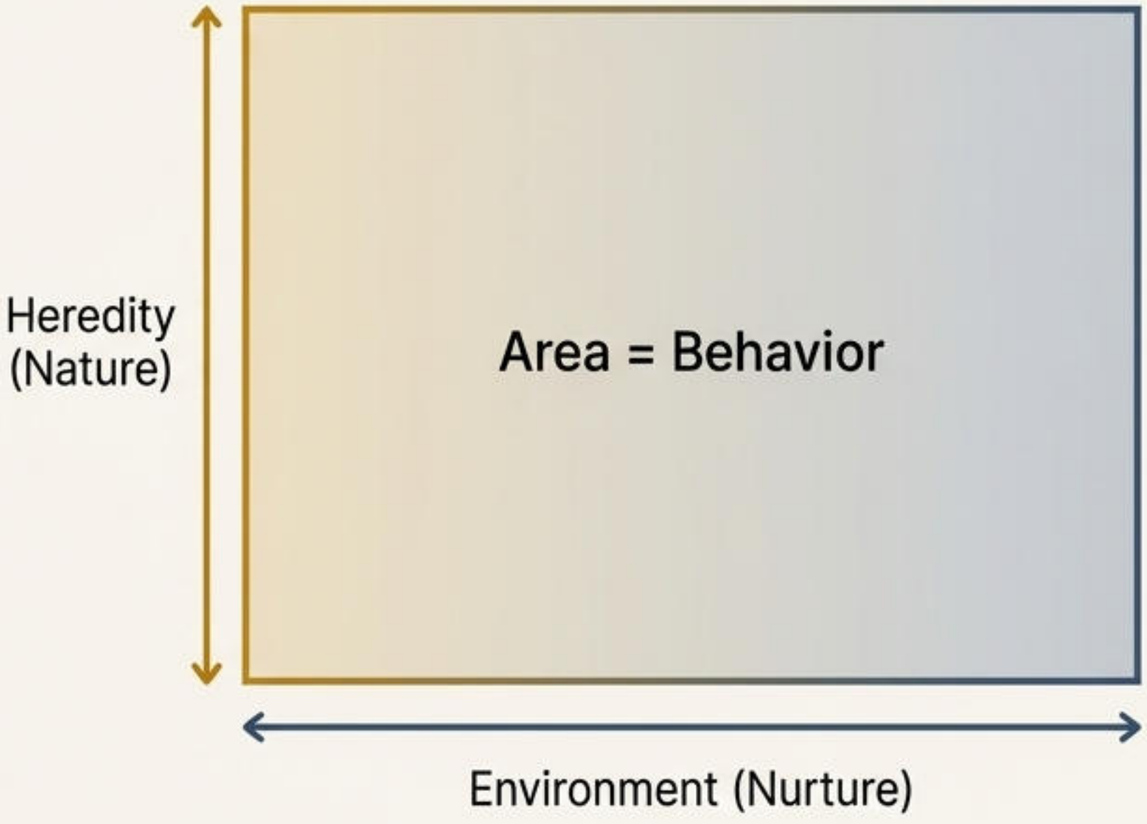
**Area = Length × Width**

Both length and width are essential for creating the area; you cannot have one without the other.

Similarly, we can conceptualize behavior as the product of heredity and environment:

**Behavior = Heredity × Environment**

Just as length and width both contribute to the area, heredity (nature) and environment (nurture) both contribute to motivating a specific behavior.

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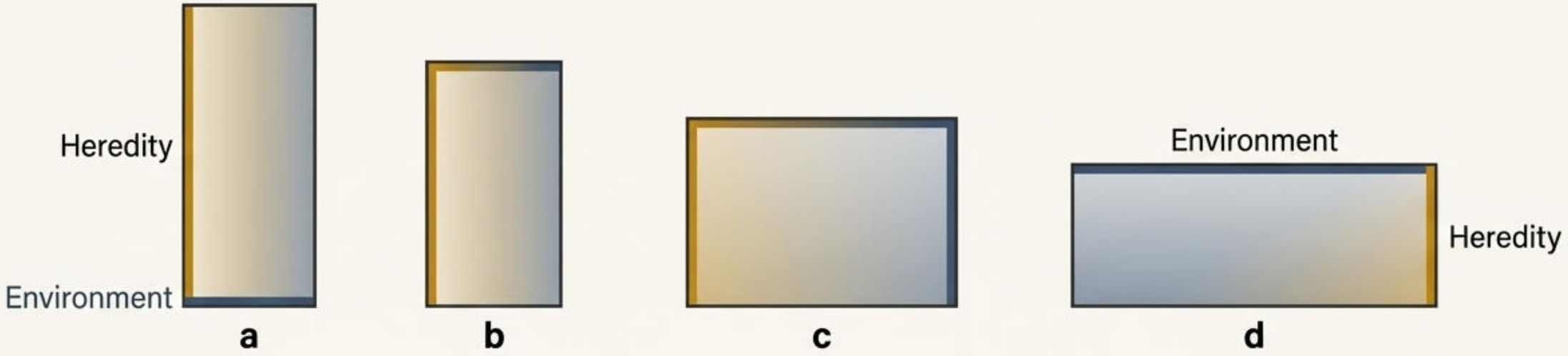
# Visualizing Relative Contributions

The rectangle metaphor also illustrates how the *relative* contributions of nature and nurture can vary for different behaviors.




- In **rectangle a**, length (heredity) is the dominant contributor to the area.
- As we move towards **rectangle d**, the contribution of length (heredity) decreases, while the contribution of width (environment) increases.

Crucially, in every single rectangle, *both* length and width are present and necessary to create the area. Likewise, for any given behavior, both heredity and environment are always contributing factors, even if their relative influence changes.

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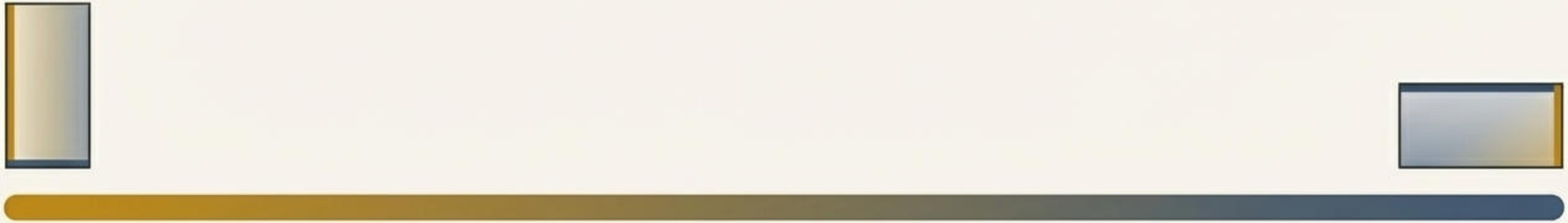
# A Spectrum of Influence: From Innate to Learned

This model helps us understand why some behaviors appear more ‘built-in’ than others.



### Genetically Disposed Behaviors (Innate)

These are strongly influenced by heredity and require very little environmental experience to emerge. Learning to eat is a primary example.

### Genetically Neutral Behaviors (Learned)

These require significant environmental experience and practice to occur. Mastering a complex skill like long division falls into this this category.

Most behaviors lie somewhere between these two extremes, involving a dynamic interplay of genetic predisposition and environmental input.

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# Understanding Innateness, Part 1: Inherited Physical Features

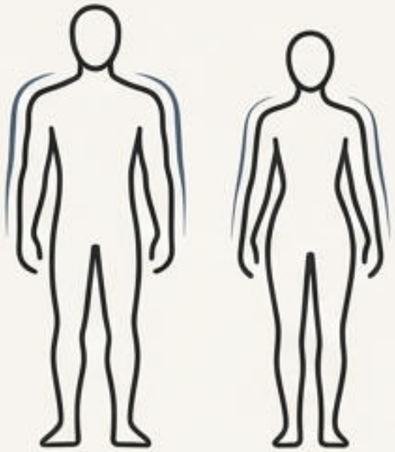
Heredity influences behavior in two primary ways. The first is through inherited physical features that enable or constrain certain actions.

## Anatomy

Consider the relationship between the thumb and fingers. The dexterity afforded by this arrangement is a direct result of our evolved anatomy and is fundamental to complex tasks like writing or tool use.

## Physical Dimorphism

On average, human males are larger and have greater muscle mass than females. These inherited physical differences have historically contributed to general divisions of labor (e.g., heavy work like farming vs. lighter work like clerical tasks), demonstrating how a physical inheritance can shape behavioral patterns.



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# Understanding Innateness, Part 2: Inherited Dispositions

The second, more subtle way heredity influences behavior is through the innateness of certain feelings, motives, or emotional reactions. These are not predetermined behaviors but rather dispositions to react in one way over another to specific stimuli.

The influence of these innate dispositions is apparent in the universal prevalence of:

- **Sexual Behavior:** The fundamental drive is biologically rooted.
- **Specific Fears:** A predisposition to fear stimuli that were ancestrally dangerous (e.g., snakes, loud noises).
- **Taste Preferences:** A baby's innate preference for sweet tastes and aversion to bitter ones.
- **Aesthetic Pleasures:** The seemingly universal human enjoyment of music.

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# The Indispensable Role of Experience

It is critical to understand that “innate” does not mean “independent of environment.” As the rectangle metaphor shows, even behaviors that are heavily influenced by heredity (rectangle a) still require some minimal environmental experience to occur.

- **Walking:** While humans have an innate predisposition to walk, the behavior requires prior experience in sitting upright and crawling to emerge in a timely manner (Dennis, 1960).
- **Crying:** An infant’s cry to signal hunger or distress is remarkably complete at birth, requiring little practice.
- **Taste Reactions:** Facial expressions indicating pleasure or disgust at different tastes are present within hours of birth, needing no training (Steiner, 1977).

In most complex behaviors, like gender role development, innate dispositions and personal experience operate in tandem from the very beginning.

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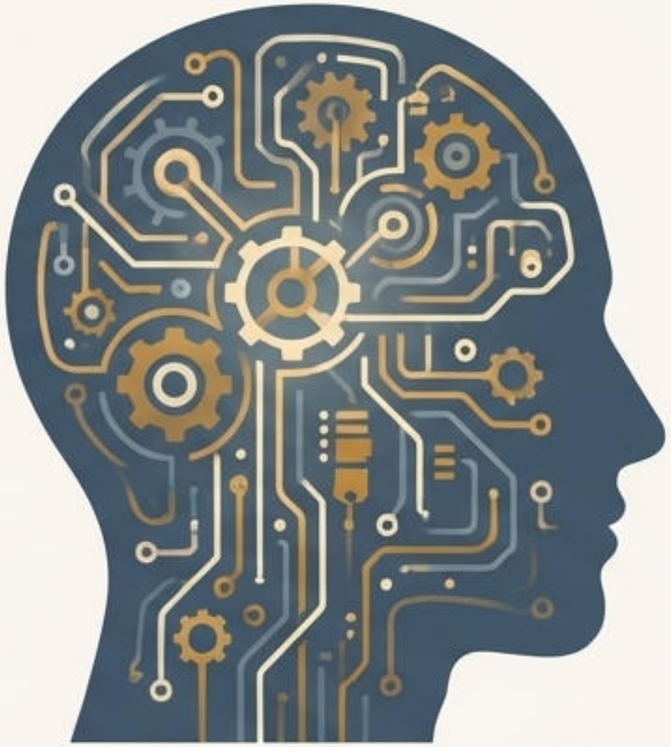
# Evolutionary Psychology

## The Mind as an Evolved Organ

A central tenet of evolutionary psychology is that human minds are not “blank slates” at birth, passively waiting for experience to write upon them.



Blank Slate

Instead, the human mind itself has been shaped by evolution through natural selection. Our evolutionary past interacts with our personal experiences to determine current motivation. The mind is an active **organ** equipped with mechanisms designed to solve specific, recurring problems of adaptation that our ancestors faced over thousands of generations.

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# Psychological Mechanisms: Evolved Solutions to Adaptive Problems

Evolutionary psychology posits that the mind contains a large number of evolved **psychological mechanisms**. These are specific, information-processing solutions that evolved to solve particular adaptive problems (Buss, 2015).

Universal motives are a type of psychological mechanism. They evolved because they motivated behaviors that enhanced survival and reproduction.

## Fear of Snakes

## Food Preferences

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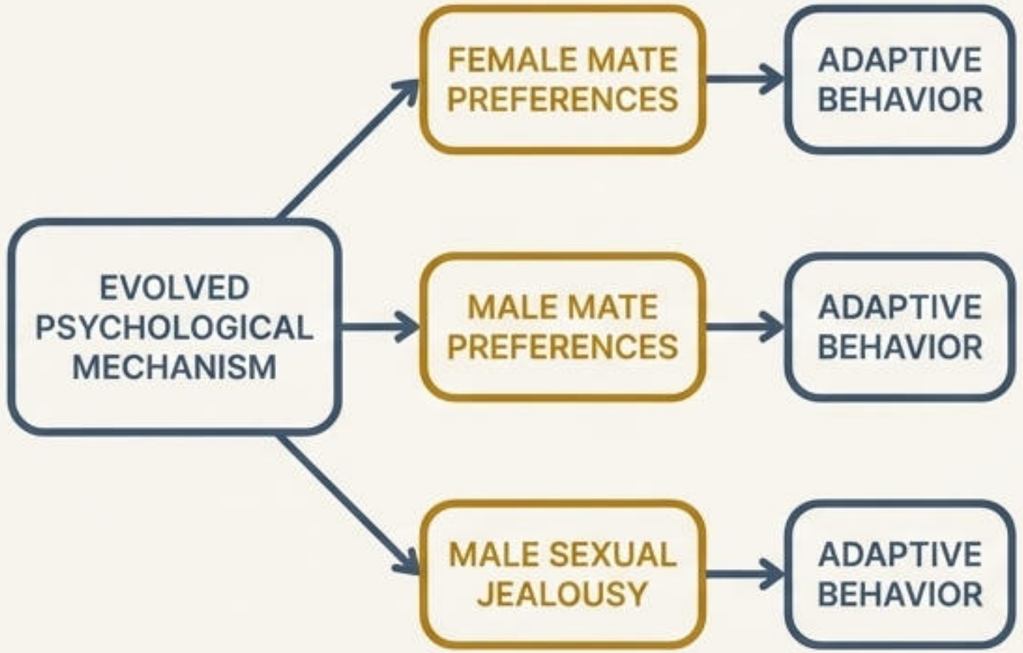
# Evolved Mechanisms in Complex Social Behavior

Psychological mechanisms also evolved to solve complex social and reproductive challenges, shaping our interpersonal motivations.

- **Female Mate Preferences:** A preference for mates with economic resources and status evolved to help ensure the provision and protection of offspring.
- **Male Mate Preferences:** An attraction to youth and beauty (as cues to fertility) and a desire for sexual variety evolved to motivate seeking fertile mates and increasing reproductive opportunities.
- **Male Sexual Jealousy:** This powerful emotion evolved as a psychological mechanism to increase a man's confidence in the paternity of his children, guarding against cuckoldry.




Each of these motives can be understood as an evolved psychological tool that aided adaptation.

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# Universal Motives

## Identifying Cross-Cultural Commonalities

If psychological mechanisms are a product of our shared evolutionary history, we should expect to find evidence of them in all human populations. A motive is considered **universal** if it occurs in all countries and cultures, even if its specific expression varies.

Despite immense differences in geography, climate, and social customs, the underlying motivations remain the same. Donald E. Brown (1991) uses the term **universals** to describe traits found in almost all cultures and societies. These universals provide a window into our shared, evolved human nature.

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# Categories of Universal Motives, Part 1

Brown's list of universals highlights several key categories of motives and valued incentives. The sheer breadth of these universals is remarkable, showing how deeply evolutionary pressures have shaped our psychology.

## Aesthetics

Art, music, standards of beauty, and concepts of hygiene.

## Control of Environment

The use of fire, tools, shelter, and mood-altering substances.

## Emotions

A core set of emotions are universally recognized: anger, contempt, disgust, fear, happiness, sadness, and surprise.

## Facial Expressions

Universal facial expressions exist for these core emotions, serving as a fundamental communication system.

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# Categories of Universal Motives, Part 2

The list of universals is dominated by social and survival-oriented **motives**, reflecting the primary challenges of our evolutionary past.

- **Fears:** Common fears of ancestrally relevant dangers like snakes, loud noises, and strangers in childhood.
- **Goal Setting:** The cognitive ability to plan and predict the future to alleviate stress.
- **Self-Concept:** A universal understanding of the self as a distinct subject and object, different from others.
- **Sexual Interactions:** Universal patterns of attraction, sexual jealousy, and social regulation of sexual behavior.
- **Social Milieu:** All humans live in social units with established rights, obligations, and concepts of status.

Entire social institutions have developed to satisfy and regulate these universal motives.

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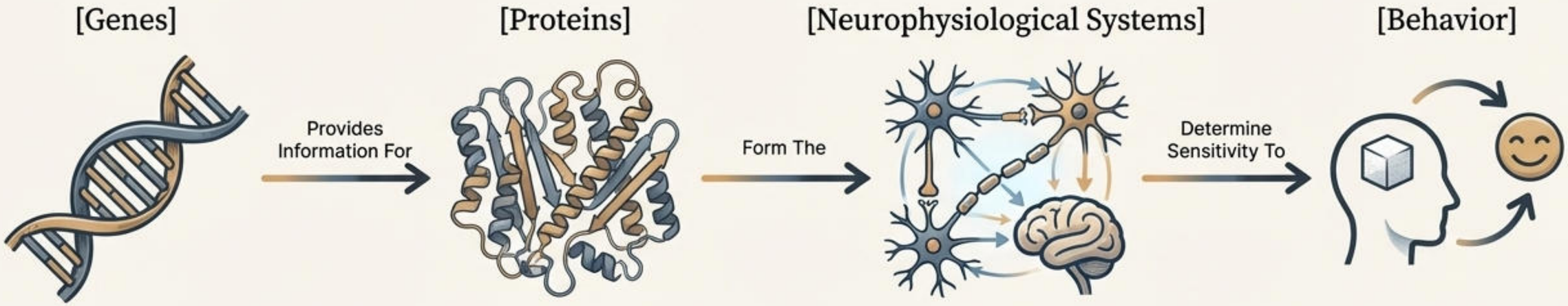
# Genetic Foundations

## The Biological Transmission of Motives

How are these universal motives passed down through generations? The mechanism is **genetic transmission via natural selection**. Genes, segments of DNA, provide the information necessary for constructing proteins. These proteins, in turn, form the body's physiological systems, including the brain and nervous system.

It is crucial to understand that genes do not influence behavior directly. Rather, they provide the recipe for building neurophysiological systems that determine our sensitivity and reaction to environmental stimuli (Plomin et al., 2013). The inheritance of a motive means inheriting a brain and body that are particularly sensitive to the stimuli that evoke or satisfy that motive. For instance, genes code for the tongue's taste receptors, which create an infant's preference for sweet tastes—a visible behavior whose roots are genetic.

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